

Software Requirements Specification (SRS): AI Fitness Coach

1. Project Overview

The AI Fitness Coach is a web-based platform designed to provide hyper-personalized fitness and nutrition guidance. By leveraging AI, the system translates user-specific physical data and goals into actionable daily workout plans and nutritional insights, while providing tools for tracking long-term progress.

2. Executive Summary

The platform aims to reduce the barrier to entry for professional fitness coaching by automating plan generation. Users provide biological data and equipment availability; the system responds with structured fitness programs, dietary suggestions, and data visualization tools to foster habit consistency and physical improvement.

3. Stakeholders

- **End Users:** Individuals seeking personalized fitness and nutrition guidance.
- **System Administrators:** Staff responsible for monitoring platform health and user management.
- **AI Model Providers:** Third-party services providing the generative intelligence for workouts/nutrition.

4. Target Users

- **Beginners:** Users needing guidance on form and routine structure.
- **Intermediate/Advanced:** Users seeking periodization or variety in their training.
- **Health-Conscious Individuals:** Users tracking weight and body measurements for health management.

5. Functional Requirements

- **User Onboarding:** System shall capture age, gender, height, weight, fitness level, goals, and equipment availability.

- **AI Plan Generation:** System shall generate daily workout routines and meal recommendations based on user profiles.
- **Workout Tracking:** Users shall be able to log completed exercises, sets, reps, and duration.
- **Progress Monitoring:** System shall provide a dashboard to visualize weight and body measurement changes over time via charts.
- **Daily AI Guidance:** System shall push a daily motivational or educational fitness tip to the user dashboard.
- **Notification Engine:** System shall send email notifications for upcoming scheduled workouts and weekly progress summaries.
- **Authentication:** System shall support secure account creation, login, and password management.

6. Non-Functional Requirements

- **Availability:** The system shall maintain 99.9% uptime.
- **Security:** All user data (PII) must be encrypted at rest and in transit.
- **Responsiveness:** The UI must render correctly on desktop, tablet, and mobile browsers.
- **Scalability:** The system shall support concurrent requests from 10,000+ users.
- **Performance:** AI-generated plan responses should be delivered within 5 seconds.

7. User Stories

- **As a user**, I want to input my equipment list so that I don't get exercises I cannot perform.
- **As a user**, I want to track my weekly weight trends to see if I am meeting my goals.
- **As a user**, I want to receive a weekly email summary so that I stay motivated and informed.
- **As a user**, I want my workout plan to adjust as I mark exercises as completed or too difficult.

8. Acceptance Criteria

- **Onboarding:** Profile is successfully saved to the database upon completion.
- **Workout Generation:** AI returns a valid routine that matches the user's specified equipment.
- **Progress Charts:** Data points must match the logs entered by the user.
- **Emails:** Notification service must trigger at the designated user time or weekly interval.

9. Business Rules

- Users must be at least 18 years of age.
- Workout plans must be generated based on the specific "Fitness Level" category selected by the user.
- Measurements (weight/body stats) can only be updated once per 24 hours to prevent data entry errors.

10. Assumptions

- Users have access to reliable internet connectivity.
- The chosen AI model provider has sufficient uptime for routine generation.
- Users are responsible for their own physical safety; the platform assumes a "use at your own risk" policy.

11. Constraints

- The application must adhere to GDPR/CCPA data privacy standards.
- The AI must not provide medical or clinical nutritional advice (requires disclaimer).
- No integration with third-party wearable devices (e.g., Apple Health, Fitbit) in the initial release.

12. Risks

- **AI Hallucination:** Risk of the model suggesting unsafe movements or improper nutritional advice.
- **Data Leakage:** Unauthorized access to sensitive user biometric data.
- **Integration Failure:** Failure of the third-party AI API could render the core service unavailable.

13. Out of Scope

- Live video coaching or real-time form correction via camera.
- Social media integration or "community" features (leaderboards/forums).
- Mobile native applications (iOS/Android).
- E-commerce storefront for supplements or equipment.

14. Open Questions

- Which AI model (e.g., GPT-4, Claude) is preferred for plan generation?
- What frequency of updates should the system enforce for the user's "Fitness Level"?

- Are there specific email service providers (e.g., SendGrid, AWS SES) preferred for notification delivery?